CANARA COLLEGE LIBRARY MANGALURU-3

Reg. No.

CANARA COLLEGE LIBRARY MANGALURU

BCAAIMS 203

Second Semester B.C.A. (AI & ML) Degree Examination, June/July 2026
(SEP) (2024 – 25 Batch Onwards)
COMPUTER ARCHITECTURE

Time : 3 Hours Max. Marks : 80

***Note** : Answer **any ten** questions from Part – **A** and **one full** question from **each** Unit of Part – **B**.*

PART – A **(10×2=20)**

1. a) Obtain the 1's and 2's Complements of $(1110.101)_2$.

   b) Write BCD and 84-2-1 code of $(98)_{10}$.

   c) Perform :

      i) $(1000011)_2 + (1101101)_2$

      ii) $(101110111)_2 - (11011101)_2$

   d) Write truth table and logic expression of NOR gate.

   e) Define :

      i) Pair

      ii) QUAD.

   f) How to write the complement of a Boolean function ? Also write the complement of F(X, Y, Z) = X'YZ' + X'Y'Z.

   g) What is Combination logical Circuit ? Draw the block diagram.

   h) What is Half Subtractor ? Write its truth table.

   i) What is Multiplexer ?

   j) What is a counter ? Minimum of how many flip-flops are required to design 3-bit counter ?

   k) Draw the circuit diagram of SR latch using NOR gate.

   l) What is Flip flop ? Write the excitation tables of D flip flop.

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PART – B (4×15=60)

Unit – I

2. a) Perform following conversion. (5+6+4)
   i) $(1101.11)_2 = (\ )_{10}$
   ii) $(37)_8 = (\ )_{16}$
   iii) $(45)_{10} = (\ )_2$

   b) Perform following Subtraction using 1's and 2's complement methods.
   i) $(1001)_2 - (1011)_2$
   ii) $(10011)_2 - (1001)_2$.

   c) State any four Postulates of Boolean algebra.

3. a) Perform following conversion. (5+6+4)
   i) $(111101.10)_2 = (\ )_8$
   ii) $(45.B3)_{16} = (\ )_2$
   iii) $(789)_{10} = (\ )_2$

   b) Perform following Subtraction using 9's and 10's complement methods.
   i) $(6408)_{10} - (543)_{10}$
   ii) $(272)_{10} - (354)_{10}$.

   c) State and prove any two theorems of Boolean algebra.

Unit – II

4. a) Express the Boolean function F(A, B, C) = A + B'C as sum of minterm and product of maxterm. (5+5+5)

   b) Using K-Map simplify the following expression.
   F(A, B, C, D) = Σ(0, 3, 4, 7, 8) + Σd(10, 11, 12, 13, 14, 15).

   c) Prove that NAND is universal gate.

5. a) Implement Boolean function F = x'y'z + x'yz + yz' with basic gate and also write the truth table. (5+5+5)

   b) Using K-Map simplify the following expression
   F(w, x, y, z) = Σ(8, 9, 10, 11, 12, 13, 14).

   c) Implement the Boolean function F(A, B, C) = A + B'C using only NOR gate.

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Unit – III

6. a) Explain the working of Full Adder. **(5+5+5)**
   b) Design BCD to Excess-3 code converter.
   c) Explain the working of 2 to 4 line Decoder.

7. a) Design and explain BCD Adder. **(6+4+5)**
   b) Design Octal to Binary Encoder.
   c) Design 1 bit Magnitude comparator.

Unit – IV

8. a) Explain the working of clocked SR flip flop. **(5+5+5)**
   b) Design a 3-bit counter using JK flip flop.
   c) What are shift registers ? Explain the 4-bit shift register with the block diagram.

9. a) Explain the working of JK flip flop. **(5+5+5)**
   b) Design MOD 6 counter using T flip flop.
   c) Explain the working of Bi-directional shift register.
